

QUANTUM COMPUTING · BEGINNER NOTES

A Different Kind of Bit

Understanding why quantum information is different from classical information

09 A Different Kind of Bit

We already know that ordinary computers use **bits** to store and process information.

A quantum computer uses a different basic unit of information: the **qubit**, short for *quantum bit*.

The qubit follows the rules of quantum mechanics, which allows it to behave differently from an ordinary classical bit.

1. Start with an ordinary bit

A classical bit has two possible values:



At any particular moment, a classical bit is either **0** or **1**.

Think of a light switch:

OFF can represent 0 and ON can represent 1. The switch has one definite state at a time.

2. Now meet the qubit

A qubit also has two basic states:



These are called the **computational basis states**.

But unlike a classical bit, a qubit can also exist in a **superposition** of these two states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here:

Symbol	Meaning
$ \psi\rangle$	The overall quantum state
$ 0\rangle$	One basis state
$ 1\rangle$	The other basis state
α	Amplitude associated with $ 0\rangle$
β	Amplitude associated with $ 1\rangle$

3. What does superposition mean?

Superposition means that a qubit can be described as a combination of its basis states.

For example:

$$|\psi\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

This state is called $|+\rangle$.

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

If we measure this qubit in the computational basis, we get:

$$P(0) = 50\% \quad P(1) = 50\%$$

Important:

Saying that a qubit is in superposition does not mean that a measurement displays both 0 and 1 at the same time.

The measurement produces one classical result, while the quantum state before measurement can contain both basis-state components.

4. The qubit is not simply a "fuzzy bit"

It is tempting to think:

"The qubit is secretly either 0 or 1, but we just don't know which."

That is not the correct picture of a general quantum state.

A superposition is a genuine quantum state described by **amplitudes**. These amplitudes determine the probabilities of measurement outcomes.

5. Amplitude is not probability

This is one of the most important ideas to remember.

If:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

then the measurement probabilities are:

$$P(0) = |\alpha|^2$$

$$P(1) = |\beta|^2$$

The probabilities must add up to 1:

$$|\alpha|^2 + |\beta|^2 = 1$$

6. A simple example

Consider:

$$|\psi\rangle = \sqrt{3}/2 |0\rangle + 1/2 |1\rangle$$

The amplitudes are:

$$\alpha = \sqrt{3}/2 \quad \beta = 1/2$$

Square their magnitudes:

$$P(0) = |\sqrt{3}/2|^2 = 3/4 = 75\%$$

$$P(1) = |1/2|^2 = 1/4 = 25\%$$

So this qubit is more likely to produce 0 than 1 when measured.

7. A qubit can be changed using quantum gates

Just as classical computers perform operations on bits, quantum computers perform operations on qubits using **quantum gates**.

For example, the X gate flips the computational basis states:



$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

8. H creates a different kind of state

The Hadamard gate, H, can take a definite state and create superposition.



$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

9. The key difference

Classical Bit

Possible values: 0 or 1

State: One definite value at a time

Operations: Classical logic operations

Measurement: Reveals the bit's value

Qubit

Basis states: $|0\rangle$ and $|1\rangle$

State: Can be in a superposition

Operations: Quantum gates

Measurement: Produces a classical result

10. A useful mental picture

A classical bit is like a coin lying on a table: **heads or tails**.

A qubit is described by a quantum state that can contain contributions from both basis states.

Classical: 0 or 1

Quantum: $\alpha|0\rangle + \beta|1\rangle$

After measurement: 0 or 1

Key idea

A qubit is not just a smaller version of a classical bit. Its state can be a superposition of $|0\rangle$ and $|1\rangle$, and quantum gates can manipulate that state before measurement.

10 Bits and Qubits

Now that we know what a qubit is, let's directly compare it with the classical bit.

Understanding this difference is important because quantum computing is not simply "normal computing with faster bits."

1. One bit

One classical bit can store one of two possible values:

0

1

Therefore, one bit has **2 possible states**.

2. One qubit

One qubit also has two computational basis states:

 $|0\rangle$ $|1\rangle$

But it can also be in a superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

This gives a much richer set of possible quantum states.

3. Two bits

If we have two classical bits, each bit is either 0 or 1. Therefore the possible combinations are:

00 01 10 11

There are:

$$2^2 = 4 \text{ possible states}$$

Bit 1	Bit 2	Combined value
0	0	00
0	1	01
1	0	10
1	1	11

4. Two qubits

Two qubits have the four computational basis states:

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$

A general two-qubit state can be written as:

$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$

This means that the quantum state can contain amplitudes associated with all four basis states.

But remember:

When we measure the two qubits in the computational basis, we still obtain only one of the four classical outcomes:

$00, 01, 10, \text{ or } 11$

5. The number of basis states grows quickly

For **n classical bits**, there are:

2^n possible bit strings

The same number of computational-basis states exists for **n qubits**.

Number of bits / qubits	Number of basis states
1	2
2	4
3	8
4	16
5	32
10	1,024

6. So what's actually different?

A common misunderstanding is:

"10 qubits simply store 1,024 classical values at once."

That's not quite correct.

A quantum computer can represent a state that is a combination of many computational-basis states, with an amplitude attached to each one.

However, when we measure the system, we obtain a single classical outcome.

7. Why superposition matters

Consider one qubit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A quantum algorithm can manipulate the amplitudes α and β using quantum gates.

The goal is not simply to "look at every answer at once." Instead, quantum algorithms use operations such as **superposition**, **interference**, and sometimes **entanglement** to shape the probabilities of useful outcomes.

8. Bits vs qubits: side-by-side

Feature	Classical Bit	Qubit
Basic states	0, 1	$ 0\rangle, 1\rangle$
Can form superposition?	No	Yes
State description	0 or 1	$\alpha 0\rangle + \beta 1\rangle$
Operations	Logic gates	Quantum gates
Measurement	Reads the value	Produces a probabilistic classical outcome
Typical symbols	0 / 1	$ 0\rangle / 1\rangle$

9. Example: bit vs qubit operation

Classical bit

Start with:

0

Apply NOT:

 $0 \rightarrow 1$

The result is definitely 1.

Qubit

Start with:

 $|0\rangle$

Apply H:

 $|0\rangle \rightarrow |+\rangle$

Measurement gives 0 or 1 with equal probability.

10. The measurement bridge

Quantum information becomes classical information when we measure it.



For a single qubit measured in the computational basis:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

↓ **measurement**

0 or 1

11. A complete example

Start with a classical-looking quantum state:

$|0\rangle$

Apply the Hadamard gate:

$$H|0\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

Measure the qubit:

$$P(0) = 50\%$$

$$P(1) = 50\%$$

If we repeat the experiment many times, the actual counts will generally approach a roughly equal number of 0s and 1s.

12. What makes quantum computing powerful?

Qubits provide a different way of representing and manipulating information.

Quantum algorithms can take advantage of:

Superposition — quantum states can combine basis states.

Interference — amplitudes can reinforce or cancel each other.

Entanglement — multiple qubits can have correlations that have no classical equivalent.

These ideas work together to create algorithms that can outperform classical approaches for certain problems.

Key idea

A classical bit stores a definite 0 or 1. A qubit uses a quantum state that can be a superposition of $|0\rangle$ and $|1\rangle$. Quantum gates manipulate these states, and measurement converts the quantum information into classical results.

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Quick Recap

Remember	Meaning
Bit	Classical unit of information: 0 or 1
Qubit	Quantum unit of information
Superposition	A quantum state can combine $ 0\rangle$ and $ 1\rangle$
Amplitude	Complex number describing a component of the quantum state
Probability	Obtained from the squared magnitude of an amplitude
Measurement	Produces a classical outcome
Quantum gate	Changes a qubit's quantum state